



Minia University

Faculty of Engineering

Mechatronics & Industrial Robotics Programm



Electronic Circuits Report Proteus Simulation Tasks

Course:

Electronic Circuits

Submitted to:

Dr. / Mona Sayed Abdelkarim

Prepared By:

- | | |
|------------------|------------------|
| 1. Muhammed Nasr | 4. Gehad Marwan |
| 2. Gerges Morzaq | 5. Mohrael Attia |
| 3. Habiba Ashraf | |

Academic Year

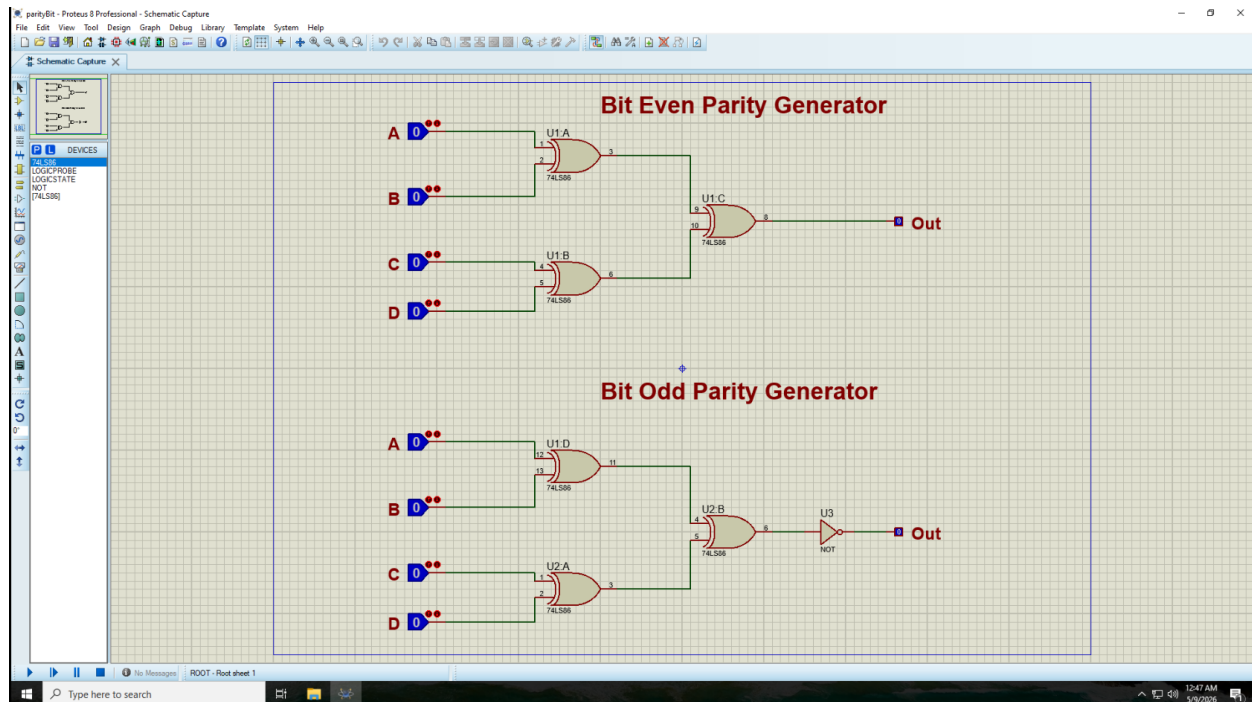
2025 – 2026

Table of Contents

Task 01: Parity Bit Circuit.....	3
Task 02: Half Adder Circuit	4
Task 03: Full Adder Circuit	5
Task 04: Sub adder Circuit and Applications	6
Task 05: 8-Bit Adder Using 74HC283	7
Task 06: 4-Bit Comparator from Scratch	8
Task 07: 2-Bit Comparator Circuit.....	9
Task 08: Decoder Circuit	10
Task 09: Encoder Circuit.....	11
Task 10: Multiplexer Circuit	12
Task 11: Demultiplexer Circuit.....	13
Task 12: S-R Latch Circuit.....	14
Task 13: R-S Latch Circuit.....	15
Task 14: D Latch Circuit.....	16
Task 15: J-K Latch Circuit.....	17

Task 01: Parity Bit Circuit

Muhammed



Objective:

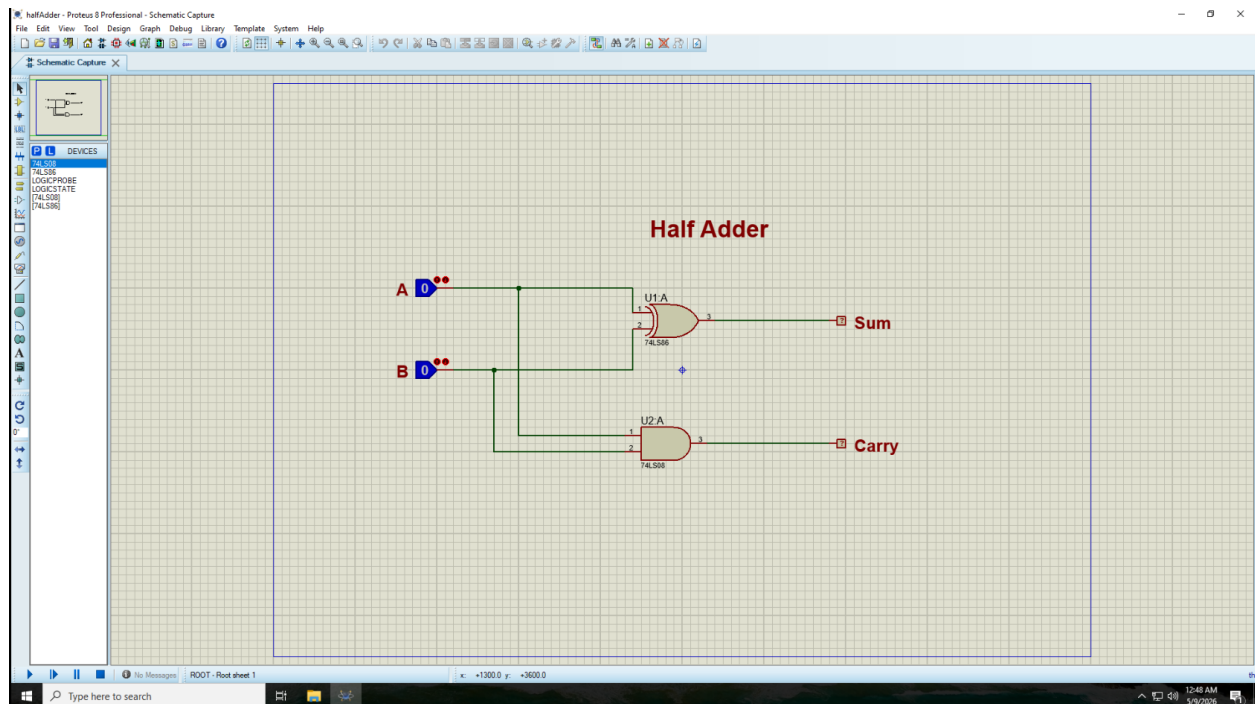
To design and simulate a 4-input parity bit generator using logic gates.

Theory:

A parity bit is used for error detection in digital communication. In even parity, the total number of 1s must be even.

Task 02: Half Adder Circuit

Muhammed



Objective:

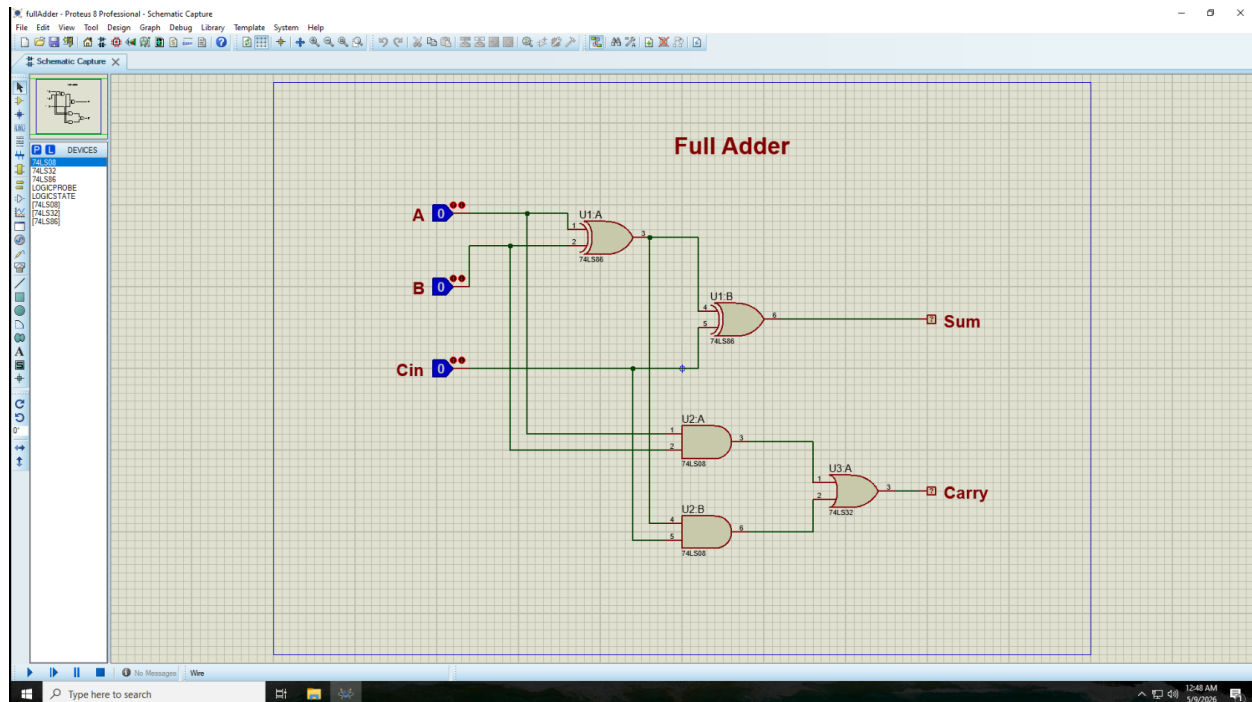
To design and simulate a half adder circuit using logic gates.

Theory:

A half adder is a combinational logic circuit used to add two single-bit binary numbers. It produces two outputs: Sum and Carry.

Task 03: Full Adder Circuit

Muhammed



Objective:

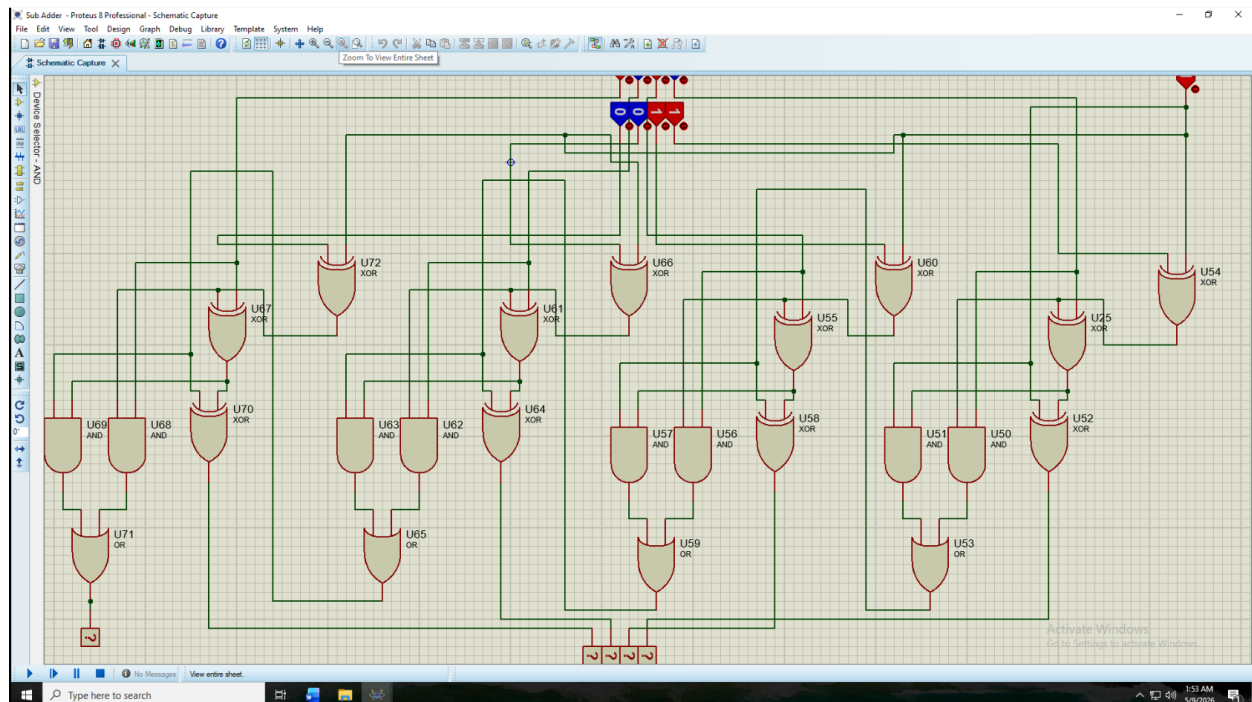
To design and simulate a full adder circuit using logic gates.

Theory:

A full adder is a combinational logic circuit used to add three binary inputs: A, B, and Cin. It produces two outputs: Sum and Cout.

Task 04: Sub adder Circuit and Applications

Gerges



Objective:

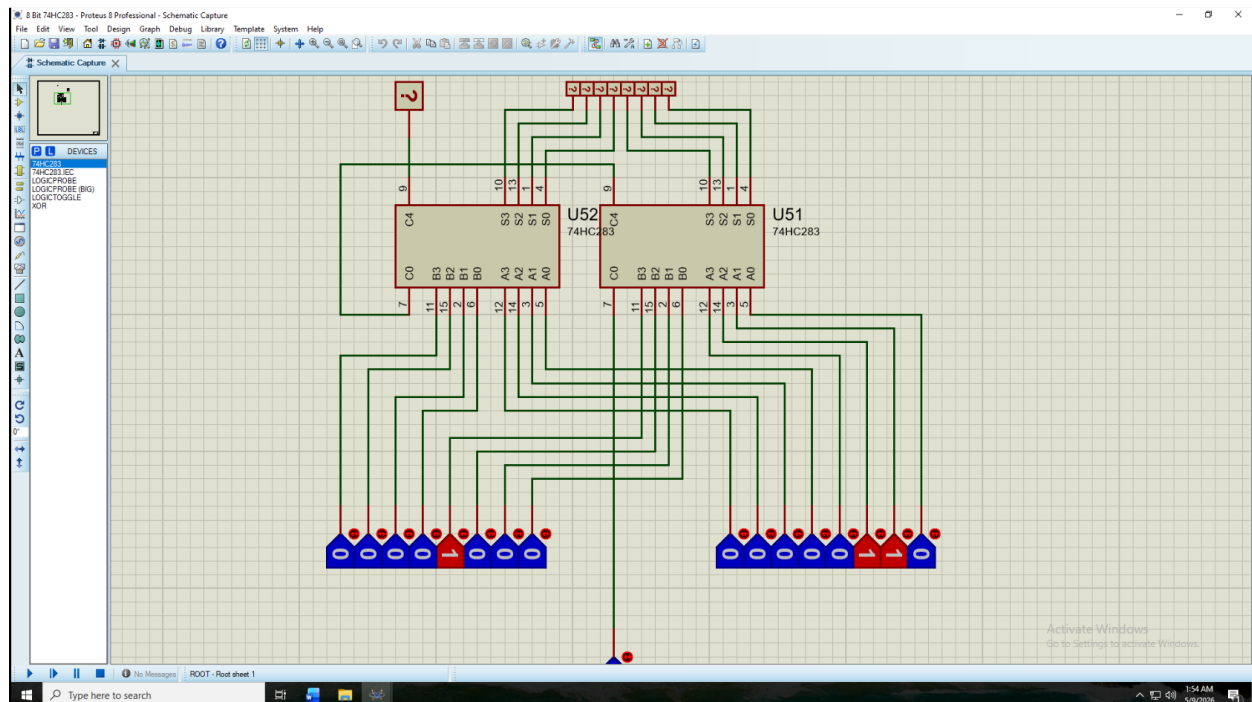
To design and simulate a 4-bit adder-subtractor circuit using basic logic gates.

Theory:

The circuit performs addition when Mode = 0 and subtraction when Mode = 1 using the 2's complement method.

Task 05: 8-Bit Adder Using 74HC283

Gerges



Objective:

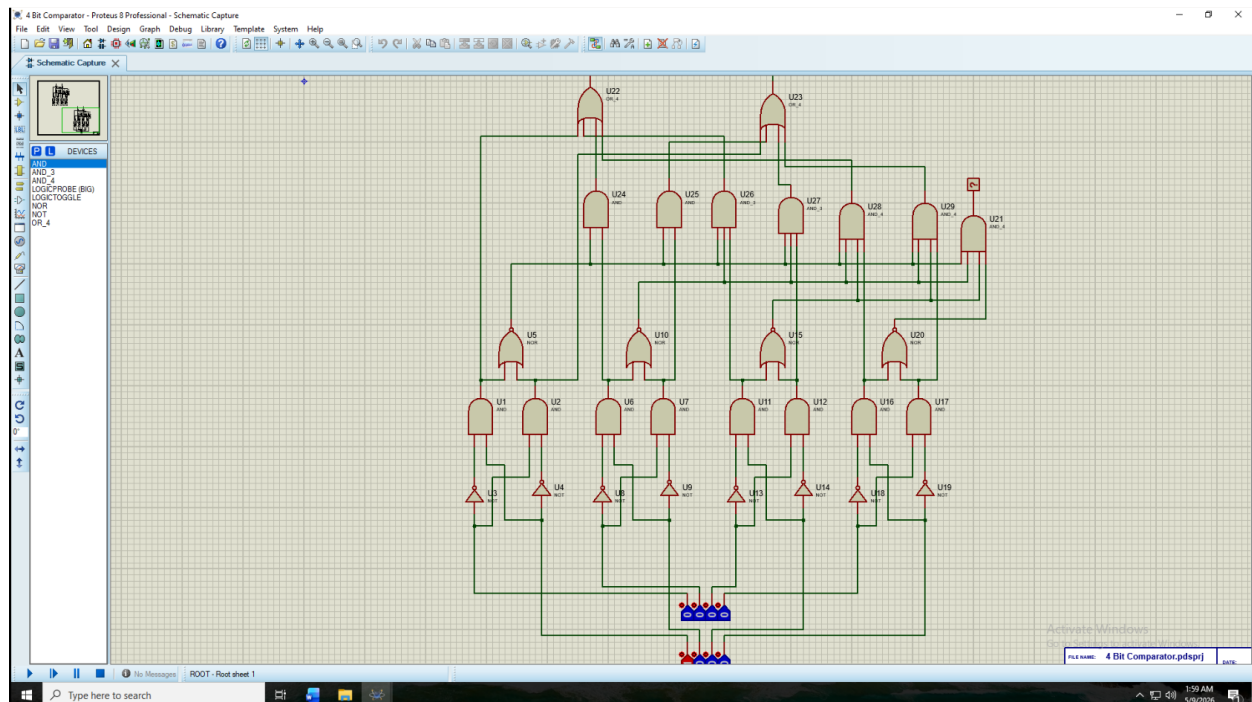
To design and simulate an 8-bit binary adder using two 74HC283 4-bit adder ICs.

Theory:

The 8-bit addition is achieved by cascading two 4-bit 74HC283 adders, where the carry output of the lower 4-bit adder is connected to the carry input of the higher 4-bit adder.

Task 06: 4-Bit Comparator from Scratch

Gerges



Objective:

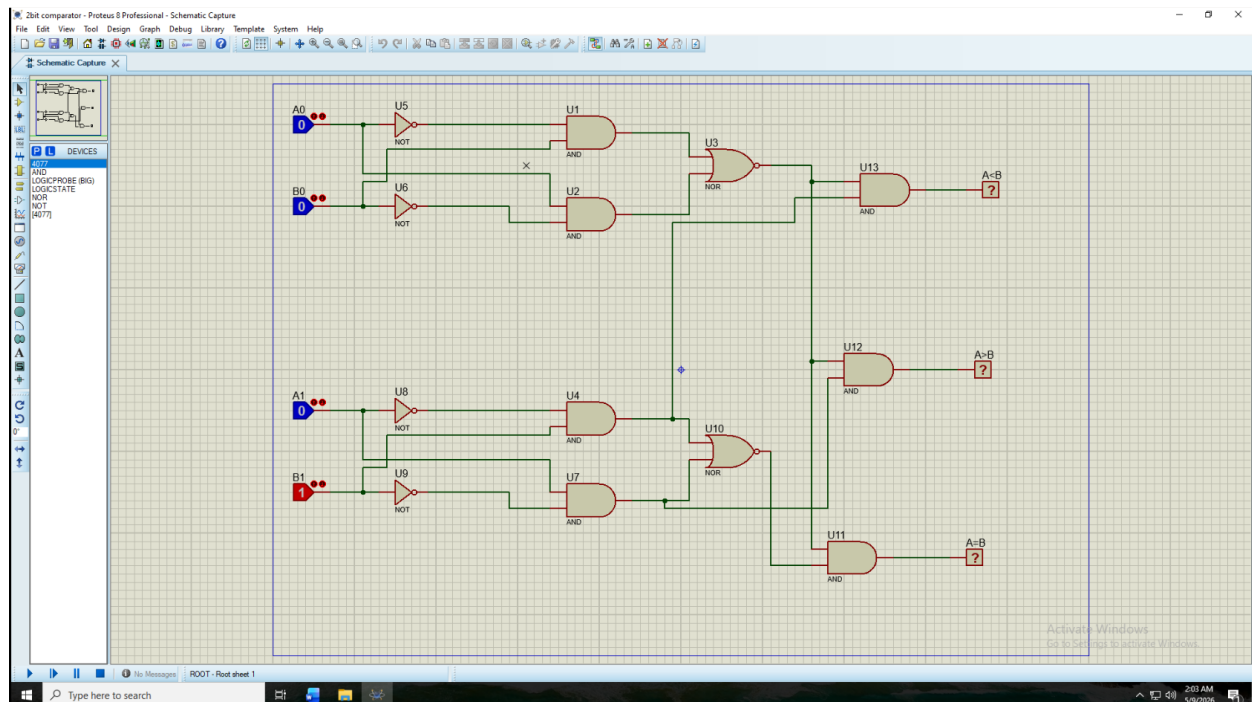
To design and simulate a 4-bit magnitude comparator using basic logic gates.

Theory:

The circuit compares two 4-bit binary numbers and determines whether A is greater than, equal to, or less than B.

Task 07: 2-Bit Comparator Circuit

Habiba



Objective:

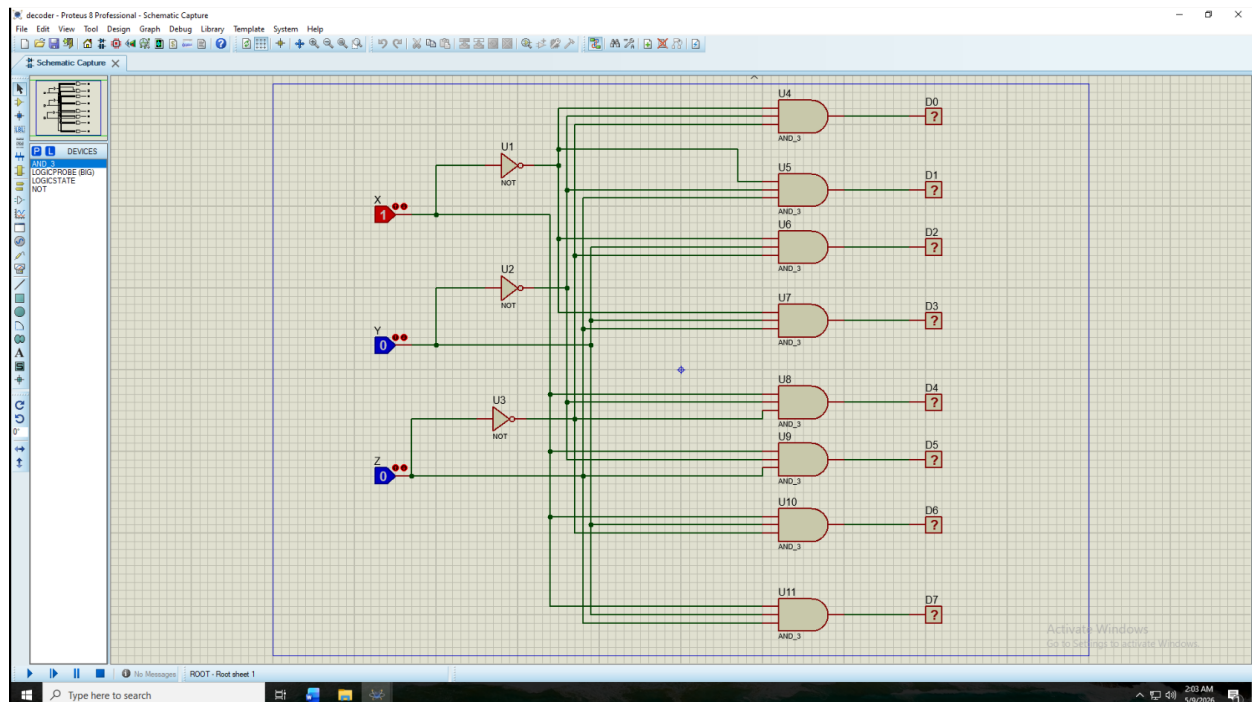
To design and simulate a 2-bit comparator circuit using basic logic gates.

Theory:

The circuit compares two 2-bit binary numbers and gives one of three outputs: $A > B$, $A = B$, or $A < B$.

Task 08: Decoder Circuit

Habiba



Objective:

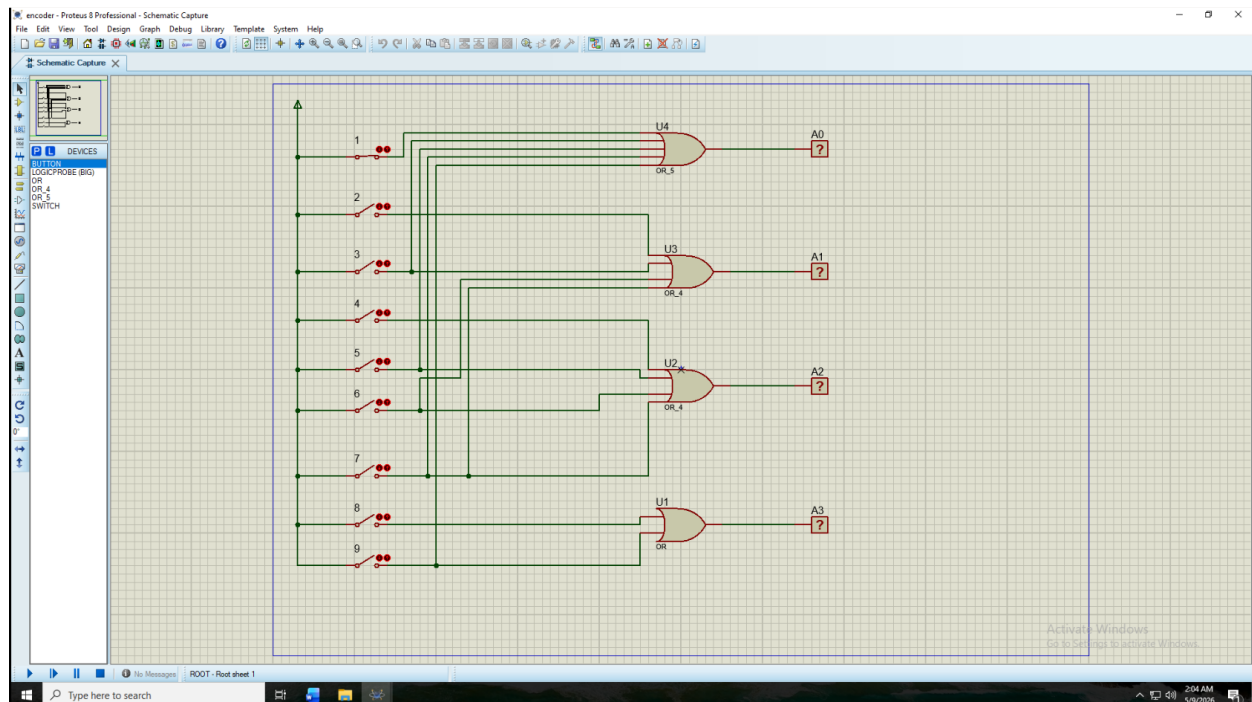
To design and simulate a decoder circuit using basic logic gates.

Theory:

A decoder converts binary input lines into one active output line based on the input combination.

Task 09: Encoder Circuit

Habiba



Objective:

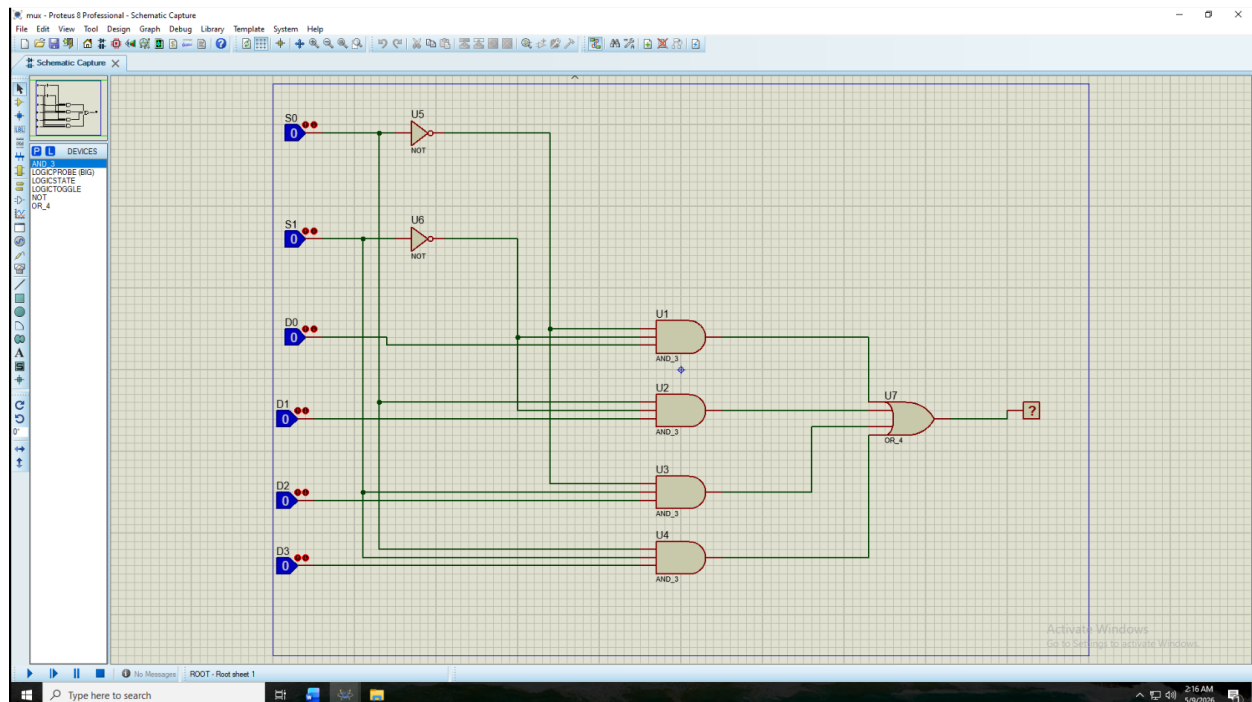
To design and simulate an encoder circuit using basic logic gates.

Theory:

An encoder converts one active input line into a binary output code that represents the active input.

Task 10: Multiplexer Circuit

Gehad



Objective:

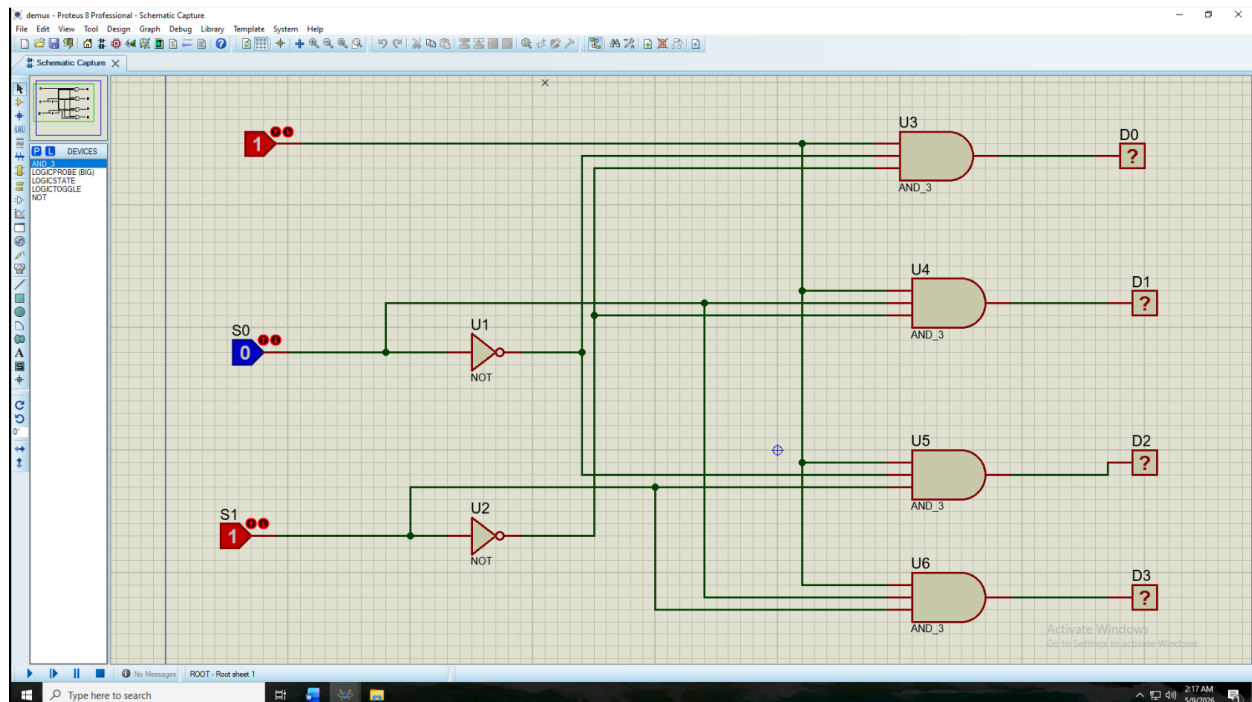
To design and simulate a multiplexer circuit using basic logic gates.

Theory:

A multiplexer selects one input from several input lines and passes it to a single output according to the select lines.

Task 11: Demultiplexer Circuit

Gehad



Objective:

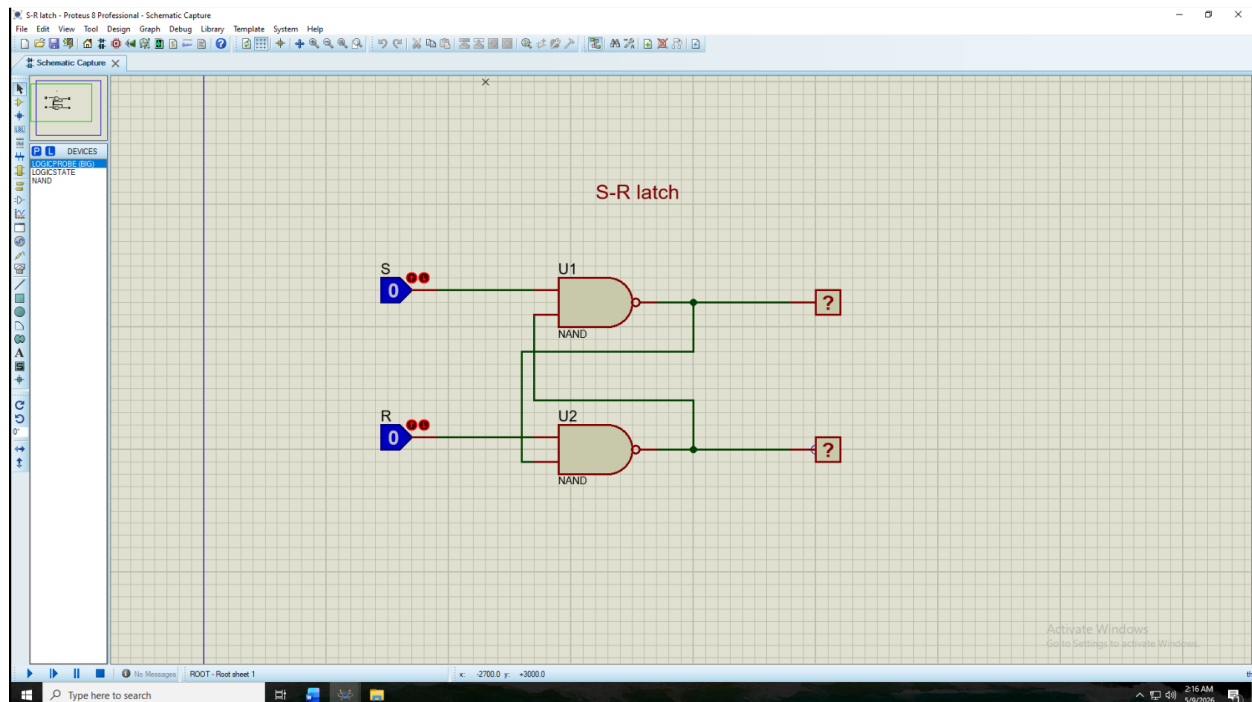
To design and simulate a demultiplexer circuit using basic logic gates.

Theory:

A demultiplexer takes one input signal and sends it to one of several output lines according to the select lines.

Task 12: S-R Latch Circuit

Gehad



Objective:

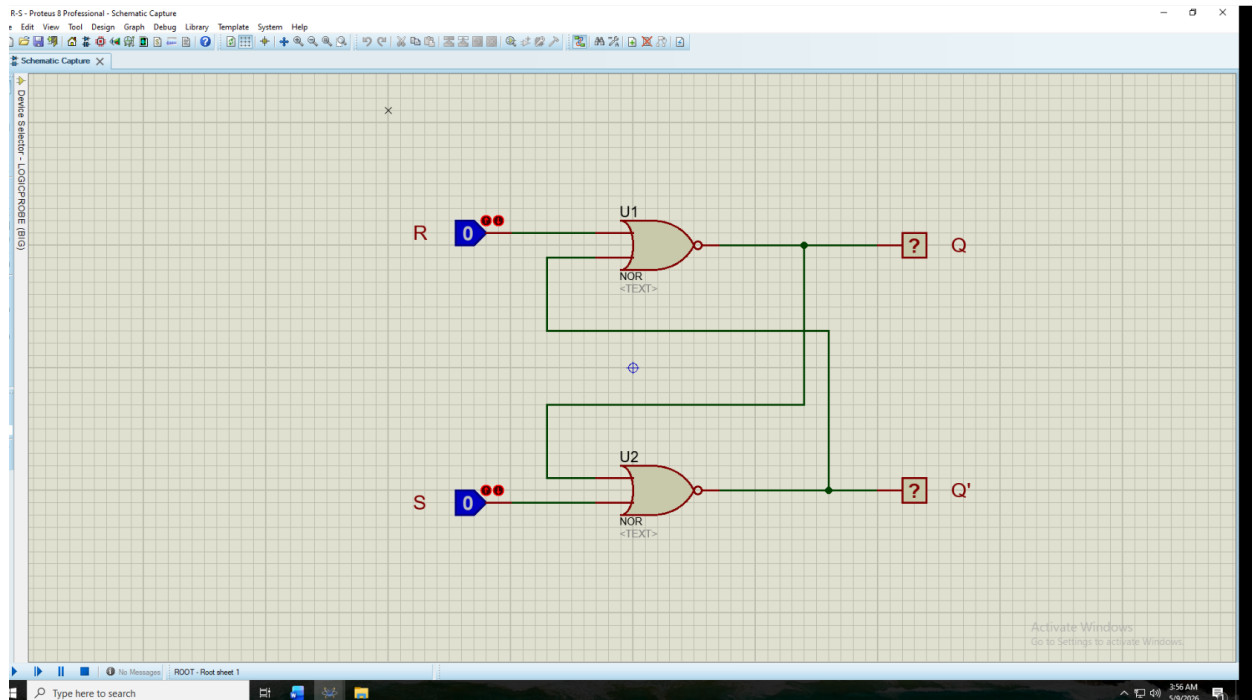
To design and simulate an S-R latch circuit using NAND gates.

Theory:

An S-R latch is a basic memory circuit that stores one bit of data using Set and Reset inputs.

Task 13: R-S Latch Circuit

Mohrael



Objective:

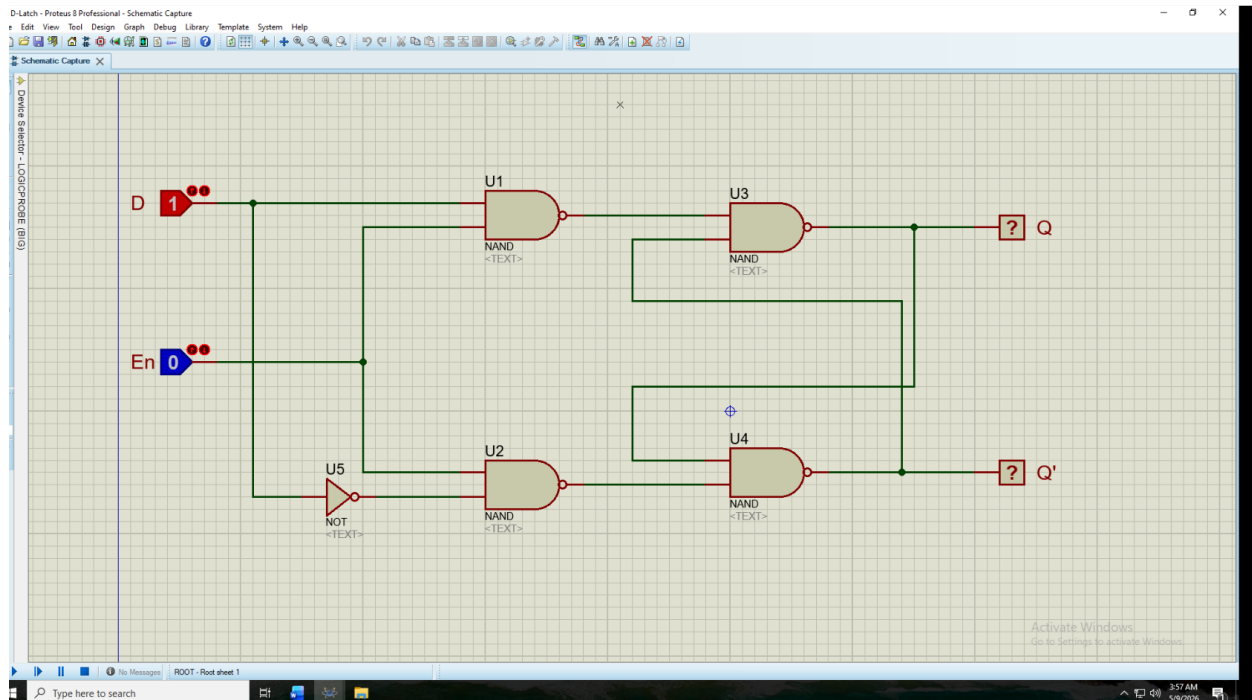
To design and simulate an R-S latch circuit using basic logic gates.

Theory:

An R-S latch is a basic memory circuit that stores one bit of data using Reset and Set inputs.

Task 14: D Latch Circuit

Mohrael



Objective:

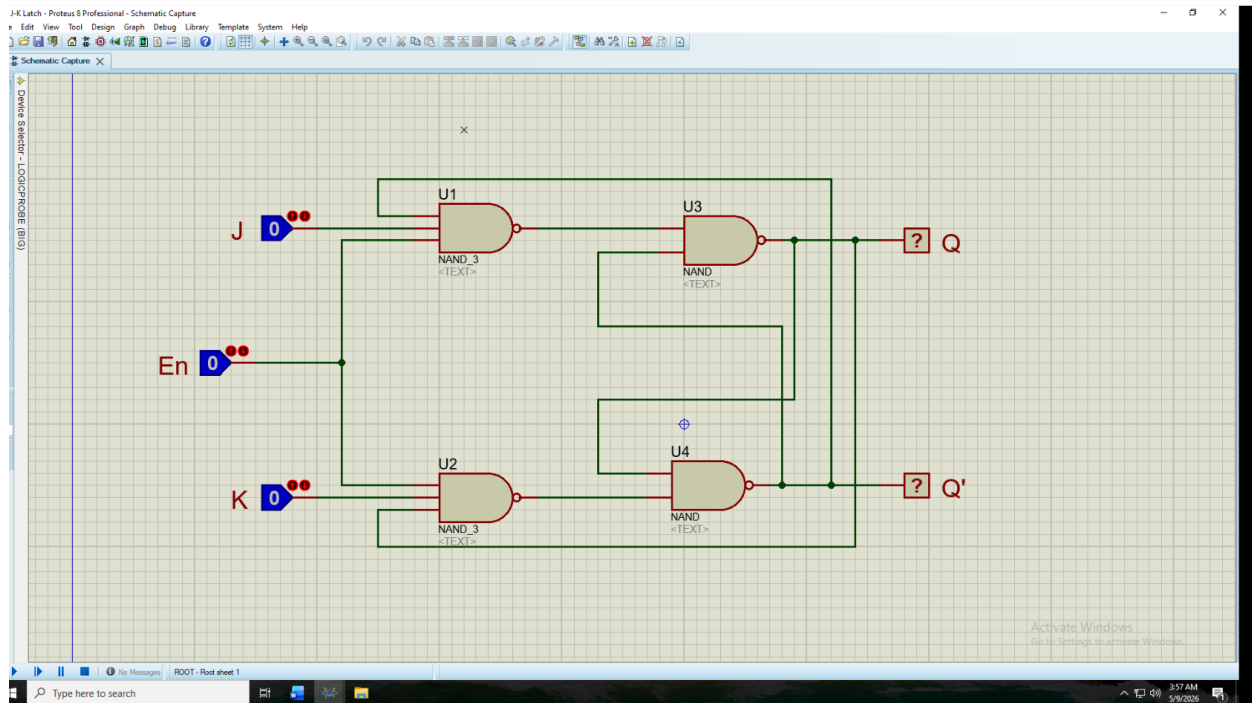
To design and simulate a D latch circuit using basic logic gates.

Theory:

A D latch stores the value of the data input D when the enable input is active.

Task 15: J-K Latch Circuit

Mohrael



Objective:

To design and simulate a J-K latch circuit using basic logic gates.

Theory:

A J-K latch is an improved latch that controls the output using J and K inputs and avoids the invalid state of the S-R latch.